

Yu Liu

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Maarintie 8, 02150 Espoo, Finland

EDUCATION

University of Chinese Academy of Sciences

M.S. in Artificial Intelligence, Hangzhou Institute for Advanced Study

Sep. 2023 – Jun. 2026 (expected)

GPA: 3.76 / 4.00

Civil Aviation University of China

B.S. in Transportation

Sep. 2015 – Jun. 2019

GPA: 3.36 / 4.00

RESEARCH & ENGINEERING PROJECTS

Multimodal Empathetic Dialogue System for Social-AI Interaction

2024 – 2026

- Designed & developed a **persona-aware multimodal agent** integrating vision, audio, and text to address the limitations of text-only LLMs in social-AI interaction; the agent can instantiate vivid, controllable virtual dialogue avatars for social scenarios. Rigorously benchmarked against SOTA (GPT-4, Gemini, DeepSeek, Qwen) via human studies, achieving **superior perceived empathy**.
- Deployed the **MER**(*Multimodal Emotion Recognition*) module as the primary function of a separate fatigue-detection system.
- Led the **visual-processing and vision-text-fusion pipelines**, implementing cross-modal alignment and dynamic emotion modeling; extended from frame-level dynamics to **multimodal ERC** (*Emotion Recognition in Conversation*) via local-global fusion and **emotion-hotspot centering**; designed multimodal fusion strategies for **expressive talking-head generation**.
- Culminated in: (i) MER—an *IEEE Transactions on Affective Computing* paper and a granted patent; (ii) talking-head generation—a *Pattern Recognition* paper and a granted patent; and (iii) two conference papers on ERC and OV-MER, respectively, under review.

Zhejiang Vanguard Project: Digital Therapeutics for Depression Detection

2024 – 2026

- Developed a **visual biomarker extraction** system for depression detection in **low-resource settings**, integrating vision models with LLM-augmented questionnaire data to enhance visual-text synergy.
- Led the **multimodal fusion design**, creating the HOPE hierarchical framework to extend prior cross-modal alignment methods for more robust visual-language integration.
- Achieved **1st Place (94.51% accuracy)** in the [ACM MM 2025 Multimodal Personality-aware Depression Detection Grand Challenge \(Young Adult Track\)](#), outperforming the 2nd place by 1.4% and the baseline by 49.14%. Served as lead author of the paper accepted to *ACM MM 2025*.

Data-Driven Aviation Analytics and Risk Modeling

2020 – 2023

- Developed & published a **complex network risk propagation model** (Spearman correlation + SIR dynamics) for flight operation risk propagation, resulting in a student first-author *EI-indexed journal publication*, later recognized as a *Frontrunner 5000 Top Article (2023)*.
- Built & deployed a Django+SQL automation platform for flight delay/fault analytics, achieving an **85% reduction** in report generation time and enabling **multi-department adoption** at China Southern Airlines.
- Led end-to-end development of data-driven solutions for aviation safety & operations, demonstrating adaptability in transitioning from aviation to advanced analytics.

PUBLICATIONS

*Co-first Author, †Supervisor,

1. **Yu Liu**, et al. *From Coarse to Nuanced: Cross-Modal Alignment of Fine-Grained Linguistic Cues and Visual Salient Regions for Dynamic Emotion Recognition*. *IEEE Transactions on Affective Computing*, 2026. [Paper Link](#).
2. Hanlei Shi*, **Yu Liu***, et al. *HOPE: Hierarchical Fusion for Optimized and Personality-Aware Estimation of Depression*. In *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM 2025)*. [Paper Link](#).
3. Taihao Li†, Leyuan Qu†, **Yu Liu**, et al. *A Dynamic Facial Expression Recognition Method, Electronic Device, and Computer-Readable Storage Medium*. Chinese Patent No. CN119206837B.
4. Taihao Li†, Leyuan Qu†, Hanlei Shi, **Yu Liu**, et al. *Facial Component-based Expression Editing Method, Electronic Device, and Computer-Readable Storage Medium*. Chinese Patent No. CN119152082B.

5. Haoxun Li, **Yu Liu**, et al. *EMORL-TTS: Reinforcement Learning for Fine-Grained Emotion Control in LLM-based TTS*. ICASSP 2026. [Paper Link](#).

6. Yantao Wang†, **Yu Liu**. *Flight Operation Risk Propagation Based on Complex Network*. Journal of Transportation Systems Engineering and Information Technology, 2020, 20(1): 198-205. [Paper Link](#).

7. Hanlei Shi, Leyuan Qu†, **Yu Liu**, et al. *Think-Before-Draw: Decomposing Emotion Semantics for Fine-Grained Controllable Generation of Expressive Talking Heads*. Pattern Recognition, 2026. [Paper Link](#).

8. Haoxun Li, Yuqing Sun, Hanlei Shi, **Yu Liu**, et al. *MSF-SER: Enriching Acoustic Modeling with Multi-Granularity Semantics for Speech Emotion Recognition*. ICASSP 2026. [Paper Link](#).

9. **Yu Liu***, Hanlei Shi*, et al. *Centering Emotion Hotspots: Multimodal Local-Global Fusion and Cross-Modal Alignment for Emotion Recognition in Conversations*. Under review, 2026. [arXiv:2510.08606](#).

10. **Yu Liu***, Lei Zhang*, et al. *Follow the Clues, Frame the Truth: Hybrid-evidential Deductive Reasoning in Open-Vocabulary Multimodal Emotion Recognition*. Under review. [arXiv:2603.16463](#).

ACADEMIC SERVICES

Reviewer, *Pattern Recognition* (Elsevier), 2025-present

Reviewer, *IEEE Transactions on Affective Computing*, 2026-present

PROFESSIONAL EXPERIENCE

Data Analyst, Airlines Operations Center, China Southern Airlines *Sep. 2019 – May 2023*

- Analyzed flight operation data and developed internal analytics tools with Python/MySQL, enabling real-time operational monitoring.
- Led dashboard automation using Tableau/QuickBI, reducing manual reporting time by over 70%. Initiated multiple engineering projects, including data integration platforms and automated scheduling tools, resulting in measurable improvements in operational efficiency.

Lab Coordinator & Administrator, Hangzhou Institute for Advanced Study, UCAS *Sep. 2024 – Present*

- Oversaw daily operations of the research lab, including member onboarding, activity organization, and GPU server resource management.
- Managed equipment procurement, research budget tracking, and reimbursement procedures, ensuring smooth project execution.

SKILLS SUMMARY

Core Research Skills: LLM/VLM fine-tuning (LoRA, adapter tuning), multimodal alignment (Video-LLaVA, CLIP), Chain-of-Thought reasoning, prompt engineering, affective computing, conversational agent design, model design for low-resource settings.

Programming & Frameworks: Python, PyTorch, HuggingFace Transformers, SQL, C++.

Engineering & Deployment: API integration, Django, scalable inference, full-stack pipelines, model serving, Tableau, QuickBI.

Languages: English (CEFR C1 Level: IELTS 7.0), Mandarin (native)

HONORS & AWARDS

Outstanding Master's Thesis *2026*
Hangzhou Institute for Advanced Study, UCAS

Outstanding Graduate *2026*
University of Chinese Academy of Sciences

Merit Student *2025*
Hangzhou Institute for Advanced Study, UCAS

Frontrunner 5000: Top Articles in Outstanding S&T Journals of China (2023) — *Sep. 2024*
“Flight Operation Risk Propagation Based on Complex Network”
Institute of Scientific and Technical Information of China (ISTIC)

First-Class Scholarship (University Level) *2024 – 2025*
Hangzhou Institute for Advanced Study, UCAS

Excellent Award, 2nd China Southern Airlines Innovation Challenge *Sep. 2021*
China Southern Airlines